

Can Tainted Pixels Expose Deepfake Videos?

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Abstract

Publicly-accessible face-manipulation tools have made deepfake creation accessible to non-expert users. Against these, existing defenses are mostly post-hoc, detecting only after forgery has occurred, and operating on still images rather than videos. Research is lacking in (i) the proactive protection of published facial videos against black-box manipulation tools, and in (ii) understanding its perceptual effect on human viewers. We introduce *TaintedPixels*, a proactive video-protection method built around an asymmetric visibility trade-off: the embedded watermark should remain inconspicuous in the published video but become obvious once a downstream tool manipulates the video. *TaintedPixels* injects structured periodic perturbations into the blue channel of facial regions and refines them under stripe-visibility, color-cast, and video-level LPIPS budgets, with lightweight motion-adaptive deployment. We believe *TaintedPixels* is the first proactive defense designed specifically against black-box manipulation tools rather than image-level pipelines or specific surrogate generators. Across three publicly available off-the-shelf video manipulation tools and two off-the-shelf detectors, *TaintedPixels* attains the highest forgery fake rate while keeping perturbations small (LPIPS = 0.0042). Our non-expert human study, conducted on a diverse set of 400 video stimuli spanning different lighting conditions, backgrounds, and skin tones, shows that protected source videos draw a 3.26% suspicion rate, while forgeries from protected sources are identified as fake much more often than forgeries from unprotected sources (90.72% vs. 56.71%). This validates the effectiveness of *TaintedPixels*.

1 Introduction

Recent advances in face swapping and audio-driven facial animation have made deepfake video synthesis both highly realistic and broadly accessible (Syed Abu Bakar et al. [2026], Baliah et al. [2025], Hong et al. [2025], Ki et al. [2025], Wang et al. [2025b]). These same capabilities, while useful for entertainment and virtual avatars, significantly lower the barrier to identity misuse, enabling convincing impersonation and the rapid spread of misinformation at scale (Alanazi et al. [2025], Diel et al. [2025, 2024]). The threat is no longer confined to expert-operated research systems: public implementations such as Roop, FaceFusion, and MuseTalk allow non-expert users to perform face swapping or lip-synchronized animation with minimal technical effort (Roop [2023], FaceFusion [2026], TME Lyra Lab [2025]). This poses a severe threat to public figures, whose abundant online appearances provide ready training material for high-fidelity forgeries that can be weaponized for political disinformation, financial fraud, and reputational harm (Chesney and Citron [2019]).

Defenses against this misuse fall into two broad families. *Post-hoc* approaches detect manipulated content after the fact (Yan et al. [2024], Cui et al. [2025], Cai et al. [2025], Saha et al. [2023]); they have demonstrated promising results, but they intervene only after a forgery has been disseminated. *Proactive* approaches instead modify benign media before the forgery, embedding either verifiable watermarks for provenance (Zhao et al. [2023], Wang et al. [2024], Wu et al. [2024], Xia et al. [2025]) or perturbations that disrupt the forgery model itself (Qu et al. [2024], Jeong et al. [2025], Wang et al. [2025c,a]). Compared to post-hoc methods, proactive approaches are becoming increasingly critical:

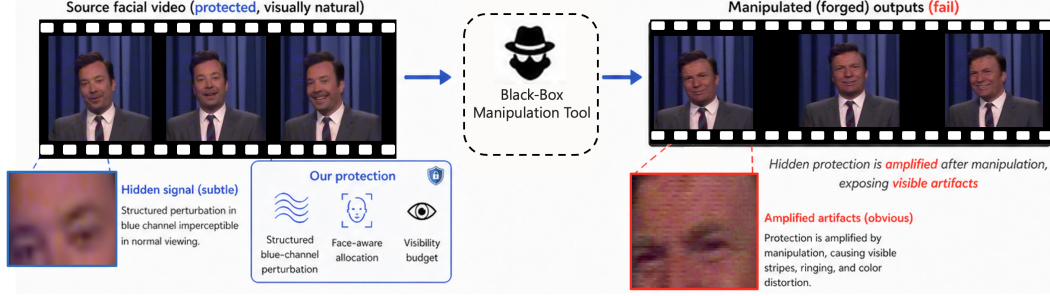


Figure 1: Inspired by anti-copy patterns in passports, *TaintedPixels* embeds an invisible signal into a source video, protecting it. Visible artifacts appear if the video is manipulated.

once a forgery spreads online, takedown is often slow and incomplete, and cannot reverse the harm already inflicted; by contrast, proactive defenses act at the source, thwarting forgeries before they can be generated and widely circulated (Nguyen-Le et al. [2025]).

Our solution. Any effective protection method must satisfy an *asymmetric visibility trade-off* with three key requirements: (i) the embedded perturbation should be imperceptible in the original video, preserving its visual quality and intended function; (ii) once processed by a manipulation pipeline, the signal should be reliably exposed to machine detectors; and (iii) the resulting forgery should appear perceptibly unconvincing or suspicious to human viewers. These objectives pull in different directions. We take a holistic view whereas previous works typically focus only on one direction.

TaintedPixels. We instantiate this principle in *TaintedPixels*, a visibility-budgeted proactive video-protection framework (Figure 1). *TaintedPixels* places structured periodic perturbations in the blue channel of facial regions, where human chromatic sensitivity is lowest (Mullen [1985], Cole et al. [1993], Curcio et al. [1991]), and refines them under three explicit budgets: a stripe-visibility budget, a color-cast budget weighted toward skin and bright regions, and a video-level LPIPS budget that bounds perceptual deviation across frames. A lightweight motion-adaptive scaling step adapts the signal to fast-moving regions without redoing optimization.

Importantly, our method requires no access to, or make any assumptions about, the downstream manipulation tool; the protection is generated solely from the source video under perceptual constraints. This design enables a broader, public-facing defense mechanism that operates at the data level, aiming not only to support detectors but also to reduce the credibility of deepfakes at scale before they can be convincingly synthesized and circulated. The perturbations from *TaintedPixels* are designed to remain imperceptible to human viewers, ensuring that the protected video preserves its visual quality and remains suitable for general use and distribution.

Threat model and scope. We envisage this scenario: a user publishes a protected video and an attacker applies an off-the-shelf manipulation tool on it, unaware that it is protected. We evaluate three representative tools spanning two manipulation paradigms: Roop (Roop [2023]) and FaceFusion (FaceFusion [2026]) for face swapping, and MuseTalk (TME Lyra Lab [2025]) for audio-driven animation. We do not claim universal robustness against future, closed-source, or adaptive forgery models. We ask the more circumscribed question of whether proactive protection can render forgeries from current public tools visibly unreliable while keeping the published video natural.

Our contributions: (i) We propose *TaintedPixels*, a novel deepfake defense which protects videos *at the source*, to facilitate detection of subsequent manipulation; (ii) we report the first-ever human perceptual study showing that protected videos, when forged, are easier to detect by the naked eye.

2 Related Work

Due to space constraints, we will not review work related to deepfake synthesis or detection. Interested readers may refer to Liu et al. [2025]. Instead, we will focus on research that is closer to ours.

Disruption-based proactive defense. Within proactive defenses, disruption-based methods perturb benign media so that downstream manipulation models fail to synthesize realistic outputs. Early work studies adversarial perturbations against facial manipulation systems (Ruiz et al. [2020]), while subsequent methods improve robustness, transferability, efficiency, or perceptual quality under transformations, compression, or specific model families (Aneja et al. [2022], Wang et al. [2022a,b], Qu et al. [2024], Zhu et al. [2025], Jeong et al. [2025], Huang et al. [2022]). Recent identity-cloaking methods further protect facial identity under black-box face swapping by perturbing identity features or weakening identity transfer (Wang et al. [2025c,a], Kung et al. [2025]).

Our work follows the disruption-based line but differs from prior work in both setting and objective. Instead of optimizing image-level perturbations against a specific surrogate generator or focusing primarily on identity suppression, we insert video-level protection against off-the-shelf black-box manipulation tools. This shifts the objective from suppressing identity transfer to exposing manipulation-induced artifacts after the downstream pipeline. Accordingly, our perturbation design emphasizes source-side perceptual budgeting and video-level robustness, rather than generic norm-bounded noise. We instantiate this idea with structured chromatic perturbations motivated by human color sensitivity (Mullen [1985], Curcio et al. [1990]) and constrain perceptual deviation using learned perceptual similarity (Zhang et al. [2018]).

Watermarking and provenance-based defenses. Another proactive approach embeds verifiable signals into facial media for authentication, attribution, provenance tracking, or proactive detection. Representative methods include recoverable provenance tags, identity watermarks, landmark perceptual watermarks, detector-aware adversarial watermarks, and contour-hybrid watermarks (Wang et al. [2021], Zhao et al. [2023], Wang et al. [2024], Wu et al. [2024], Xia et al. [2025]). These methods are useful for verification and source tracing, but they usually require a watermark extraction or detection stage. *TaintedPixels* instead treats the embedded signal as a disruption source rather than a recoverable provenance message. No watermark detector is needed.

Human perception and social media dynamics. Human perception and online dissemination jointly motivate proactive defenses. People often struggle to distinguish manipulated content, especially when it appears realistic (Diel et al. [2024], Groh et al. [2022], Köbis et al. [2024]), so a forgery that looks plausible can still mislead viewers even if detected by machines. At the same time, manipulated media spread rapidly and widely, while corrections rarely reach the same audience (Vosoughi et al. [2018], Shao et al. [2018]), meaning takedowns often come too late (Chesney and Citron [2019]). These findings motivate two design choices in *TaintedPixels*: protection should be embedded *at the source*, and its effect should appear as *visible artifacts* in forged outputs, analogous to anti-copy patterns in secure documents that only emerge upon reproduction (Garg et al. [2023]).

3 Method

3.1 Overview

We formulate proactive facial video protection as a visibility-budgeted disruption problem. Given a clean facial video $\mathbf{V} = \{x_t\}_{t=1}^T$, where each frame $x_t \in [0, 1]^{H \times W \times 3}$, our objective is to produce a protected video $\hat{\mathbf{V}} = \{\hat{x}_t\}_{t=1}^T$ that satisfies two asymmetric requirements. Before manipulation, the protected video should remain visually natural to human viewers. After manipulation, the same hidden protection should be amplified by publicly available off-the-shelf tools and make the forged output visually unreliable or easier to detect as fake. *We emphasize that our method treats the manipulation tool as a black-box.* We do not use the tool’s internals; we do not even require API access to the tool to implement our protection.

As shown in Figure 2, our framework contains three components. First, we construct a structured blue-channel perturbation prior that concentrates energy in locally textured regions. Second, we use a face-aware spatial prior to allocate more perturbation budget to regions that manipulation tools are likely to process. Finally, in the optimization and video generation stage, we optimize the perturbation under visibility budgets and generate protected videos using lightweight motion-adaptive scaling with a video-level LPIPS constraint.

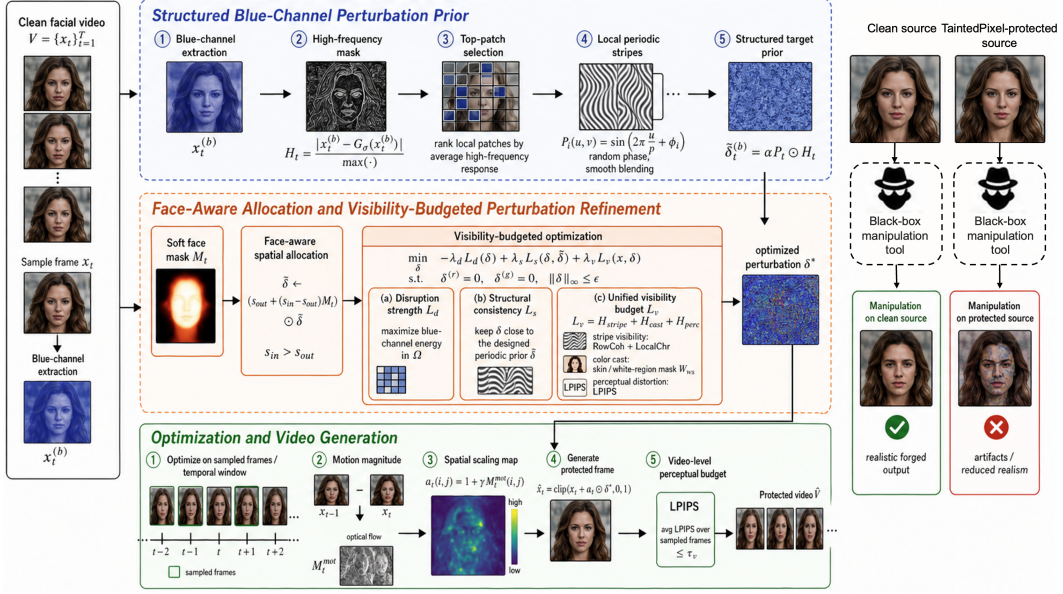


Figure 2: Overview of our proposed visibility-budgeted facial video protection framework.

3.2 Structured Blue-Channel Perturbation Prior

We restrict the perturbation to the blue channel and use $\text{clip}(z, 0, 1)$ to denote element-wise clipping of pixel values to the valid image range $[0, 1]$: $\delta_t = [0, 0, \delta_t^{(b)}]$, $\hat{x}_t = \text{clip}(x_t + \delta_t, 0, 1)$.

This design is motivated by the asymmetric sensitivity of human perception and downstream image processing. Humans are relatively less sensitive to small blue-channel variations (Mullen [1985], Cole et al. [1993], Curcio et al. [1991]), yet face manipulation pipelines still process these variations through normalization, alignment, feature extraction, and rendering (Rössler et al. [2019], Li et al. [2020a], Tolosana et al. [2020], Mirsky and Lee [2021], Walczyna and Piotrowski [2023]). Thus, a small structured chromatic signal can remain unobtrusive in the published source video but become amplified by manipulation operations.

We construct a structured target perturbation $\tilde{\delta}_t$ that combines local periodic patterns with a high-frequency content mask, so that the protection signal is preferentially placed in locally textured regions where small chromatic variations are less perceptible while still retaining disruption potential after manipulation (Mullen [1985], Ferwerda et al. [1997]): $\tilde{\delta}_t^{(b)} = \alpha P_t \odot H_t$.

Here, $H_t \in [0, 1]^{H \times W}$ is a high-frequency mask, P_t is a local periodic pattern, and $\odot$ denotes element-wise (pixel-wise) multiplication. The mask H_t is computed from the blue channel by subtracting a Gaussian-smoothed version:

$$H_t = \frac{|x_t^{(b)} - G_\sigma(x_t^{(b)})|}{\max |x_t^{(b)} - G_\sigma(x_t^{(b)})| + \xi}; \quad P_i(u, v) = \sin\left(\frac{2\pi}{p}u + \phi_i\right). \quad (1)$$

where G_σ denotes Gaussian smoothing, and ξ is a small numerical constant for stable division. We then rank local patches according to their average high-frequency response and inject periodic signals only into the top fraction of patches. This concentrates perturbations in visually complex regions, where small changes are less noticeable.

For a selected local patch, the periodic pattern, $P_i(u, v)$, is defined as shown in Eq. (1), where p is the stripe period and ϕ_i is a random phase. Overlapping patches are blended by a smooth window to avoid block boundaries. This design enforces spatial coherence in the perturbation. Compared to independent random noise, such coherent patterns are more robust to rendering transformations and are thus more likely to manifest as structured artifacts after manipulation.

3.3 Face-Aware Allocation and Visibility-Budgeted Perturbation Refinement

3.3.1 Face-Aware Allocation

Public face manipulation tools do not process all pixels equally. Face swapping and face animation models usually crop, align, normalize, and re-render the facial region, while background pixels are often ignored or weakly processed (Qu et al. [2024], Wang et al. [2025c]). Under a fixed visibility budget, perturbation energy placed in the background is therefore less useful than the perturbation energy placed on the face.

To exploit this observation, we estimate a soft face mask $M_t \in [0, 1]^{H \times W}$ and use it to reweight the target perturbation:

$$\tilde{\delta}_t^{(b)} \leftarrow (s_{\text{out}} + (s_{\text{in}} - s_{\text{out}})M_t) \odot \tilde{\delta}_t^{(b)}. \quad (2)$$

Here s_{in} and s_{out} control the perturbation scale inside and outside the face region, respectively. We use a soft mask rather than a hard one to avoid visible discontinuities along the facial boundary. In practice, the mask can be obtained from landmarks or face detectors; when landmarks are unavailable, a coarse face region is sufficient because the subsequent visibility constraints refine the perturbation.

This face-aware allocation does not assume access to the forgery model. It only encodes a weak semantic prior: facial regions are more likely to be transformed by downstream tools. This prior is particularly useful under black-box conditions, where gradients or internal features of the manipulation tool are unavailable.

3.3.2 Visibility-Budgeted Perturbation Refinement

The structured prior provides a useful initialization, but directly applying it may either be too weak to affect manipulation or too visible in the source video. We therefore refine the perturbation by solving a visibility-budgeted optimization problem. The key principle is to maximize disruption potential while penalizing only the parts of the perturbation that exceed content-adaptive perceptual budgets.

For each optimized frame (or a sampled temporal window), we solve

$$\begin{aligned} \min_{\delta} \quad & -\lambda_d \mathcal{L}_d(\delta) + \lambda_s \mathcal{L}_s(\delta, \tilde{\delta}) + \lambda_v \mathcal{L}_v(x, \delta), \\ \text{s.t.} \quad & \delta^{(r)} = 0, \quad \delta^{(g)} = 0, \quad \|\delta\|_{\infty} \leq \epsilon. \end{aligned} \quad (3)$$

Here, δ denotes the optimized perturbation and $\tilde{\delta}$ denotes the structured target perturbation constructed from the blue-channel prior. The three terms correspond to three goals: $\mathcal{L}_d$ encourages sufficient disruption strength, $\mathcal{L}_s$ preserves the designed spatial structure, and $\mathcal{L}_v$ enforces source-side visual quality. The constraints $\delta^{(r)} = \delta^{(g)} = 0$ restrict the perturbation to the blue channel, while $\|\delta\|_{\infty} \leq \epsilon$ enforces a strict magnitude bound.

Disruption term. This encourages the perturbation, $\mathcal{L}_d(\delta)$, to use the blue-channel budget:

$$\mathcal{L}_d(\delta) = \frac{1}{|\Omega|} \sum_{(i,j) \in \Omega} \left(\delta^{(b)}(i,j) \right)^2; \quad \mathcal{L}_s(\delta, \tilde{\delta}) = \frac{1}{|\Omega|} \sum_{(i,j) \in \Omega} \left(\delta^{(b)}(i,j) - \tilde{\delta}^{(b)}(i,j) \right)^2. \quad (4)$$

where Ω denotes the support region of the structured perturbation. Unlike conventional adversarial attacks that optimize against a specific model, our setting assumes no access to the downstream manipulation pipeline. Therefore, $\mathcal{L}_d$ does not target a particular model, but instead acts as a model-agnostic proxy objective. Intuitively, increasing the energy of the structured chromatic signal within Ω increases the likelihood that it will survive preprocessing and be amplified during manipulation.

Structural consistency. Keep the optimized perturbation δ close to the designed blue-channel prior $\tilde{\delta}$ via $\mathcal{L}_s$ in Eq. (4).

This term prevents the optimization from degenerating into arbitrary high-magnitude pixel noise. Without this constraint, maximizing $\mathcal{L}_d$ alone may produce perturbations that are less stable under resizing, normalization, and rendering, and may also become more visible in the source video. By preserving the local periodic structure of $\tilde{\delta}$, $\mathcal{L}_s$ encourages the optimized perturbation to remain spatially coherent while still allowing sufficient flexibility for visibility refinement.

Unified visibility budget. The visibility term constrains the source-side perceptual impact of the optimized perturbation:

$$\mathcal{L}_v(x, \delta) = \mathcal{H}_{\text{stripe}}(x, \delta) + \mathcal{H}_{\text{cast}}(x, \delta) + \mathcal{H}_{\text{perc}}(x, \delta). \quad (5)$$

It groups together the dominant visual failure modes caused by blue-channel structured perturbations: visible stripe patterns, unnatural color cast, and global perceptual distortion. Specifically, let $C_b(x)$ denote the blue-difference chroma component and $\Delta C_b = C_b(x + \delta) - C_b(x)$.

We compute a stripe visibility score

$$\mathcal{V}(x, x + \delta) = \beta_1 \text{RowCoh}(\Delta C_b) + \beta_2 \text{LocalChr}(\Delta C_b), \quad (6)$$

where $\text{RowCoh}(\cdot)$ measures row-wise coherent chroma changes and $\text{LocalChr}(\cdot)$ measures local chroma deviations. The corresponding normalized hinge loss, $\mathcal{H}_{\text{stripe}}$, is:

$$\mathcal{H}_{\text{stripe}} = \left[\frac{\mathcal{V}(x, x + \delta)}{\mathcal{B}_{\text{vis}}(x) + \xi} - 1 \right]_+; \quad \mathcal{C}(x, x + \delta) = \frac{\sum_{i,j} W_{\text{ws}}(i, j) |\Delta C_b(i, j)|}{\sum_{i,j} W_{\text{ws}}(i, j) + \xi}, \quad (7)$$

where $\mathcal{B}_{\text{vis}}(x)$ is a content-adaptive visibility budget. To suppress color cast, especially in skin and bright white regions, we define a soft white/skin mask W_{ws} and measure the masked chroma shift $\mathcal{C}(x, x + \delta)$, defined in Eq. (7). The color-cast hinge, $\mathcal{H}_{\text{cast}}$, is

$$\mathcal{H}_{\text{cast}} = \left[\frac{\mathcal{C}(x, x + \delta)}{\mathcal{B}_{\text{cast}}(x) + \xi} - 1 \right]_+; \quad \mathcal{H}_{\text{perc}} = \left[\frac{\frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \text{LPIPS}(x_t, \hat{x}_t)}{\tau_v + \xi} - 1 \right]_+, \quad (8)$$

where $\mathcal{B}_{\text{cast}}(x)$ is a content-adaptive color-cast budget.

Finally, we constrain perceptual distortion $\mathcal{H}_{\text{perc}}$ by averaging LPIPS over a temporal window $\mathcal{T}$, as per Eq. (8), where τ_v is the video-level perceptual budget. This term constrains the average perceptual deviation across frames rather than a single optimized frame. Overall, $\mathcal{L}_v$ (Eq. (5)) penalizes only the perturbation components that exceed their visibility budgets, allowing the method to preserve visual quality while still maximizing useful disruption strength.

3.4 Optimization and Video Generation

We optimize the perturbation using Adam with projected updates. After each Adam step, we project the perturbation back to the feasible blue-channel set: $\delta \leftarrow \Pi_{\|\delta\|_\infty \leq \epsilon}(\delta)$, $\delta^{(r)} = \delta^{(g)} = 0$.

For efficiency, we optimize the structured perturbation on a sampled subset of frames and propagate it to the full video. This strategy is motivated by the temporal redundancy of facial videos: adjacent frames usually share similar facial appearance, illumination, and background structure, so the same optimized protection signal can be reused across neighboring frames instead of being independently optimized for every frame.

To account for local temporal changes, we apply motion-adaptive perturbation scaling during video generation. Specifically, we estimate a normalized motion magnitude map $M_t^{\text{mot}} \in [0, 1]^{H \times W}$ between consecutive clean frames using optical-flow magnitude. We then define a spatial scaling map: $a_t(i, j) = 1 + \gamma M_t^{\text{mot}}(i, j)$, where γ is a small coefficient controlling the maximum additional perturbation strength. Given the optimized perturbation δ^* , the protected frame is generated as $\hat{x}_t = \text{clip}(x_t + a_t \odot \delta^*, 0, 1)$.

This deployment strategy reuses the same structured perturbation in relatively stable regions while assigning slightly stronger protection to fast-moving regions. The intuition is that temporal changes can mask fine-grained spatial distortions, allowing fast-moving regions to carry a small amount of additional protection without noticeably degrading source-video quality. The video-level LPIPS constraint further regularizes this process by ensuring that the average perceptual deviation across sampled frames remains within a video-level budget.

The overall procedure is practical for public-tool black-box protection because the protection is generated entirely from the published source video and perceptual constraints. It does not require gradients, intermediate features, outputs, or internal parameters from the downstream manipulation tools. The resulting protected video is then published. When an attacker applies a public manipulation tool on it, the hidden structured perturbation should be amplified by the tool’s preprocessing and rendering pipeline, leading to visible artifacts or reduced realism in the forged output.

4 Experiments

4.1 Experimental Setup

Public-tool black-box setting. We evaluate three representative off-the-shelf tools. For face swapping, we use Roop (Roop [2023]) and FaceFusion (FaceFusion [2026]). For audio-driven face animation, we use MuseTalk (TME Lyra Lab [2025]). These tools represent a realistic misuse scenario where non-expert attackers directly apply public manipulation pipelines.

Datasets and protocol. We evaluate on real facial videos sampled from FaceForensics++ (Rössler et al. [2019]) and Celeb-DF (Li et al. [2020b]), two representative datasets widely used in deepfake detection. FaceForensics++ provides 1,000 real videos and Celeb-DF provides 890 real videos. Although both datasets contain real and fake videos, we use only their real videos for the video-protection experiments. For each video, we generate both a clean input and a protected input. Each manipulation tool is then applied to both inputs, producing paired forged videos. All methods are evaluated under the same manipulation and detection pipeline.

Parameter setting. We use $\sigma = 1.0$ for the high-frequency mask, 96×96 overlapping patches with initial stride 32, and initial stripe period $p = 2$. The face mask is obtained from facial landmarks with a detector fallback and smoothed with $\sigma_M = 6.0$. We set $s_{\text{out}} = 0.8$ and select s_{in} from a small range centered at 12. The perturbation is optimized with Adam for at most 120 steps with learning rate 0.009, followed by projection to the feasible blue-channel set after each update. We tune the main strength and visibility parameters by coordinate descent on the first $N_{\text{tune}} = 6$ source frames using only source-side visibility and disruption statistics, without accessing manipulation tools, detector outputs, or human-study videos. All experiments are conducted on a single NVIDIA H100 GPU with 80GB of memory. The fake manipulation pipeline runs at 1.03 frames per second, while generating a protected video with *TaintedPixels* takes 150.37 seconds per video on average.

Metrics. For source-side quality, we report PSNR, SSIM, and video-level LPIPS between $\mathbf{V}$ and $\hat{\mathbf{V}}$. For forgery-side exposure, we report both model-based and human-study results. For model-based evaluation, because our forged videos are generated by off-the-shelf tools, including MuseTalk, Roop, and FaceFusion, they act as black-box inputs to the detectors and require generalization beyond specific forgery pipelines. Given the limited availability of public detectors with strong generalization ability and batch-evaluation support, we use two representative and relatively generalizable detectors, NPR (Tan et al. [2024]) and Deep-Fake-Detector-v2-Model (Sakthi), and report fake rate and fake confidence. For human evaluation, we report the fake rate from the human-study responses.

4.2 Protection Effectiveness

Comparisons. We evaluate *TaintedPixels* against proactive SOTA methods: DFRAP (Qu et al. [2024]), CMUA (Huang et al. [2022]), FacePoison (Zhu et al. [2025]), and FaceShield (Jeong et al. [2025]). Table 1 reports the main protection effectiveness under public black-box manipulation tools. Forgeries generated from our protected videos are more easily detected as such, compared to those protected by said SOTA methods. This shows that the hidden protection is manifested after manipulation. Although our method does not achieve the best PSNR or SSIM, it obtains a more favorable video-LPIPS trade-off, indicating that our protected videos are perceptually natural. Taken together, these results show that *TaintedPixels* preserves strong source-side perceptual quality while still causing forged outputs to fail after manipulation.

Robustness. Published videos may undergo common post-processing before being manipulated. We therefore evaluate robustness to H.264 compression and re-encoding. We apply each post-processing operation to the protected video before manipulation and report the fake rate of the resulting forged outputs. Results show that re-encoding and H.264 compression reduce the fake-detection accuracy by 3.13% and 4.27%, respectively. Nevertheless, *TaintedPixels* helps preserve the protection signal under H.264 compression and re-encoding.

Qualitative Results. We use Roop as the manipulation tool and visualize some samples. Figure 3 shows that protected videos remain visually close to their clean sources. After manipulation, however, protected videos exhibit visible failures, typically manifested as yellowish and purplish color distortions in facial regions. Compared to rival protection methods, *TaintedPixels* produces more noticeable artifacts in the videos forged from protected sources, thus making them easier to detect.

Table 1: Comparison with SOTA methods (columns) under 3 manipulation tools (rows). D1 and D2 denote NPR and Deep-Fake-Detector-v2-Model, respectively. “Clean” denotes the unprotected setting, and the remaining columns correspond to forgeries generated from protected videos. The last three rows report the visual quality of the protected videos generated from each method.

Tool	Detector	Metric	Clean	DFRAP	CMUA	FacePoison	FaceShield	<i>TaintedPixels</i> (ours)
Roop	D1 Fake	Fake Rate $\uparrow$	78.33	79.63	78.64	80.33	84.32	89.12
		Fake Conf. $\uparrow$	75.67	78.72	76.87	73.95	80.32	86.58
	D2 Fake	Fake Rate $\uparrow$	82.43	84.51	83.02	83.93	86.71	90.27
		Fake Conf. $\uparrow$	72.10	75.39	74.95	80.63	81.66	86.92
FaceFusion	D1 Fake	Fake Rate $\uparrow$	80.87	80.39	82.01	82.20	84.02	88.71
		Fake Conf. $\uparrow$	73.43	76.74	75.43	74.93	79.34	89.14
	D2 Fake	Fake Rate $\uparrow$	80.29	82.31	83.34	83.72	85.48	90.64
		Fake Conf. $\uparrow$	72.93	78.46	75.54	77.23	76.82	83.47
MuseTalk	D1 Fake	Fake Rate $\uparrow$	84.78	84.83	86.39	85.23	83.54	93.68
		Fake Conf. $\uparrow$	76.32	80.92	78.65	76.82	77.36	86.46
	D2 Fake	Fake Rate $\uparrow$	82.21	84.43	84.94	82.93	85.04	92.92
		Fake Conf. $\uparrow$	76.36	78.20	77.92	74.96	75.90	90.21
Protected videos	PSNR $\uparrow$		N/A	30.23	37.43	38.34	32.73	30.54
	SSIM $\uparrow$		N/A	0.8923	0.9342	0.9296	0.9349	0.9103
	V-LPIPS $\downarrow$		N/A	0.1264	0.0527	0.0092	0.1941	0.0042

Table 2: Component ablation of *TaintedPixels*. We report average results over Roop, FaceFusion, and MuseTalk. A better method should achieve high fake rate while maintaining low source video-LPIPS.

Periodic Prior	HF Mask	Face-aware	Visibility Budget	Motion Scaling	Video LPIPS	Avg. Fake Rate $\uparrow$	Video-LPIPS $\downarrow$
—	—	—	—	—	—	81.49	—
✓	—	—	—	—	—	86.10	0.1094
✓	✓	—	—	—	—	87.17	0.2036
✓	✓	✓	—	—	—	87.84	0.1895
✓	✓	✓	✓	—	—	88.35	0.0164
✓	✓	✓	✓	✓	—	89.62	0.0103
✓	✓	✓	✓	✓	✓	90.89	0.0042

4.3 Ablation Study

We ablate the main components of *TaintedPixels* in Table 2. The structured periodic prior tests whether coherent blue-channel patterns are more effective than unstructured noise. The high-frequency mask tests whether placing perturbations in textured regions improves imperceptibility. Face-aware allocation tests whether using the manipulation-sensitive facial region improves budget efficiency. The visibility budget tests whether the method can preserve source-side naturalness while still causing forged outputs to fail. Motion-adaptive scaling and video-level LPIPS test the contribution of the video-aware deployment strategy. Results demonstrate that structured, region-aware, and temporally adaptive perturbations are all critical to effectiveness.

4.4 Human Perceptual Study

We further evaluate perceptual effects through a human study on Amazon Mechanical Turk. Forty non-expert participants viewed a randomly shuffled set of videos from four conditions (100 videos per condition): (1) *clean source*, i.e., the videos sampled from FaceForensics++ and Celeb-DF; (2) *protected source*, i.e., the output of *TaintedPixels*; (3) *forged from clean source*, i.e., a forgery generated by applying a public tool to the original unprotected video; and (4) *forged from protected source*, i.e., a forgery generated by applying the same tool to our protected video. We selected 400 videos from our sample set as the stimuli, covering diverse lighting conditions, backgrounds, and skin tones, using Roop, FaceFusion, and MuseTalk as representative manipulation methods. Participants were asked whether each video appears suspicious or fake. For source videos, we report the fraction of affirmative responses as the *false alarm rate*; for manipulated videos, we report it as the *false detection rate*.

As shown in Tab. 3, clean source videos receive 0.8%, and protected source videos receive only 3.26% false alarms, indicating that our perturbations do not noticeably affect visual quality. In contrast,

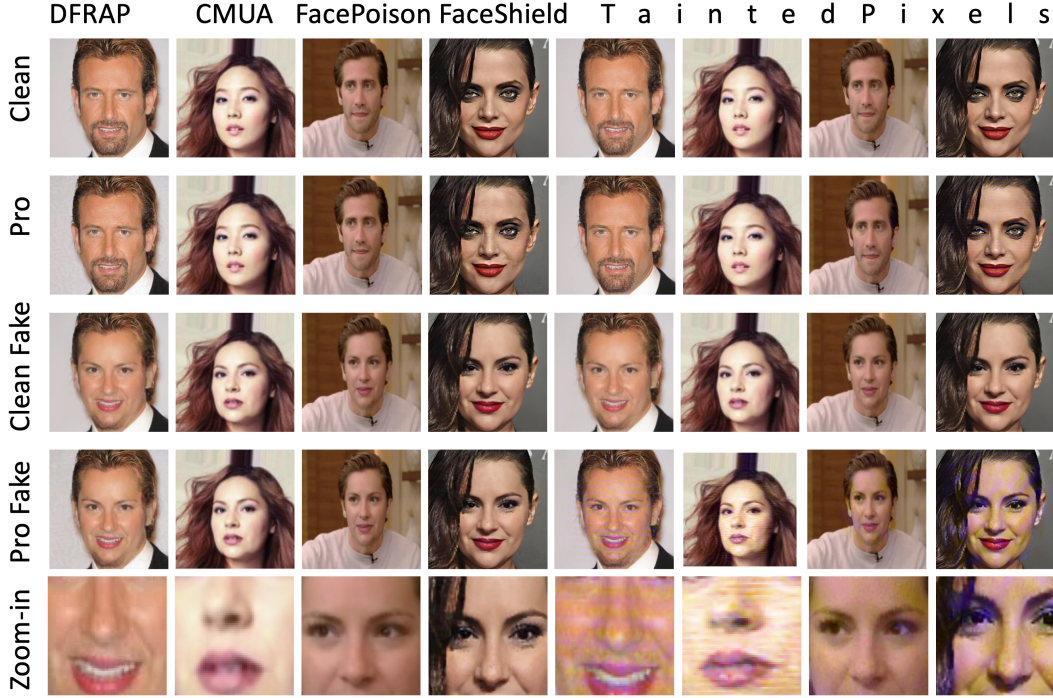


Figure 3: Qualitative results. (R1 & R2): the unprotected (clean) and protected source videos, resp. (R3): fakes generated from clean. (R4): fakes from protected. (R5): zoom in of (R4). *TaintedPixels* creates obvious purple artifacts in every video frame, easily exposing the forgery to human eyes.

Table 3: Human study results reported as the percentage of videos judged as fake. Lower values for source videos indicate better visual naturalness, while higher values for manipulated videos indicate that the forged outputs are easier for humans to recognize as fake.

Clean Source ↓	Protected Source ↓	Tool	Forged from Clean ↑	Forged from Protected ↑
0.80	3.26	Roop	54.23	89.42
		FaceFusion	51.80	94.51
		MuseTalk	64.11	88.24
		Average	56.71	90.72

forgeries from clean sources are identified as fake in 56.71% of cases on average, reflecting the baseline detectability of current manipulation tools. When the same tools are applied to protected videos, the average detection rate increases to 90.72%, a significant gain of 34.01% points. This gain demonstrates that *TaintedPixels* effectively helps humans detect forgery more readily.

5 Conclusion

To the best of our knowledge, *TaintedPixels* is the first of its kind: a proactive defense for facial videos that operationalizes an asymmetric visibility trade-off; i.e., structured periodic perturbations in the blue channel of facial regions stay inconspicuous in the published video and become exposed only after manipulation. *TaintedPixels* does not use any information about the manipulation tool, not even API access. To be sure, a sufficiently determined attacker aware of the perturbation could attempt to remove it through targeted denoising or heavy compression. We mitigate against this possibility indirectly: by keeping video-level LPIPS low, the published video is less likely to arouse suspicion even from attackers. We hope that our work encourages further research into proactive defenses.

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A Technical appendices and supplementary material

Hyperparameter and auxiliary-term definitions. To improve reproducibility and avoid ambiguity in the notation, we summarize the main initial hyperparameters used in our implementation in Table 4. We also list the auxiliary visibility-related quantities in Table 5. All experiments use the initial values in Table 4. We fine-tune the parameters based on 6 frames.

Table 4: Summary of hyperparameters.

Symbol / Term	Value / Definition	Description
α	3.5	Amplitude of the structured blue-channel prior.
r	0.8	Fraction of high-frequency patches selected.
p	2.0	Period of the local periodic pattern.
ϵ	0.25	Perturbation magnitude bound.
λ_d	3.0	Weight for $\mathcal{L}_d$.
λ_s	1.0	Weight for $\mathcal{L}_s$.
λ_v	1.0	LPIPS perceptual budget.
s_{in}	12	Perturbation scale inside the face region.
s_{out}	0.8	Perturbation scale outside the face region.
K	120	Maximum number of Adam steps.
η	0.009	Adam learning rate.

Table 5: Definitions of auxiliary visibility-related quantities.

Term	Role
$C_b(x)$	Blue-difference chroma channel used to measure blue-channel color changes.
$\mathcal{B}_{vis}(x)$	Content-adaptive stripe visibility budget estimated from clean-frame chroma statistics.
W_{ws}	Combined soft white/skin-region mask for color-cast suppression.
$\mathcal{B}_{cast}(x)$	Content-adaptive color-cast budget estimated from clean-frame white/skin chroma statistics.
$\text{RowCoh}(\Delta C_b)$	Measures row-wise coherent chroma drift, corresponding to visible stripe/banding artifacts.
$\text{LocalChr}(\Delta C_b)$	Measures local weighted blue-chroma deviation.